POSITION STATEMENT

Practical management of epidermolysis bullosa: consensus clinical position statement from the European Reference Network for Rare Skin Diseases

Abstract

Inherited epidermolysis bullosa (EB) comprises rare disorders that manifest with fragility and blistering of the skin and mucous membranes, with variable clinical severity. Management of EB is challenging due to disease rarity and complexity, the wide range of extracutaneous manifestations, and a profound impact on daily life for the patient and family members. Although reference centres providing multidisciplinary care for EB exist in each European country, it is common for healthcare professionals who are not specialized in this rare disorder to treat EB patients. Here, experts of the European Reference Network for Rare and Undiagnosed Skin Diseases (ERN-Skin, https://ern-skin.eu) propose practical recommendations for the diagnosis and management of the most common clinical issues: skin blisters and wounds, oral manifestations, pain, and itch.

Introduction

Inherited epidermolysis bullosa (EB, ORPHA:79361) represents a group of rare conditions characterized by skin and mucous membrane fragility leading to blister formation. There are four major types of EB — epidermolysis bullosa simplex (EBS), junctional EB (JEB), dystrophic EB (DEB), and Kindler EB (KEB) — and over 30 subtypes, each with variable clinical severity.

The clinical spectrum ranges widely across subtypes. Patients with localized, nonsyndromic forms typically experience blistering mainly on the hands and feet, have normal life expectancy, but may suffer significant impairment in quality of life and daily functioning. In contrast, severe subtypes can be lethal in infancy (e.g., severe JEB) or associated with multiple complications such as sepsis, failure to thrive, severe anemia, and aggressive squamous cell carcinoma, leading to reduced lifespan (e.g., severe recessive DEB [RDEB]).

Despite this heterogeneity, blisters and wounds accompanied by pain and/or itch are common features across nearly all EB subtypes.

Management of EB is complex due to its rarity, clinical variability, and multisystem involvement, which profoundly affect patients and their families. While specialized reference centres offering multidisciplinary care exist across Europe, many patients are often managed by healthcare providers outside these centres who may lack specific expertise in EB.

Comprehensive clinical practice guidelines for various aspects of EB management are available through DEBRA International (https://www.debra-international.org/copy-of-eb-health-care-patient-guid), developed using rigorous methodology based on literature review and expert appraisal. These serve as essential scientific resources.

Recognizing the need for concise, practical guidance, the European Reference Network for Rare and Undiagnosed Skin Diseases (ERN-Skin, https://ern-skin.eu) has developed this position statement to support non-specialist healthcare professionals. The aim is to provide easily implementable recommendations for managing common EB manifestations and emergency situations, enabling rapid decision-making and fostering collaboration between local providers and EB reference centres.

These consensus recommendations were developed by a panel of dermatologists, pediatric dermatologists, and dentists within ERN-Skin. They cover key areas including diagnosis, wound management, oral care, and management of pain and itch.

Part I. Diagnostic confirmation of suspected EB

Diagnosis of EB

Establishing a precise diagnosis is the first critical step in managing a patient suspected of having EB, as it provides essential prognostic information and guides long-term care.

The diagnosis of EB is based on a combination of: - Clinical findings (distribution and morphology of blisters, nail changes, dental abnormalities, etc.) - Family history (including consanguinity and inheritance pattern) - Laboratory investigations

Key diagnostic techniques include: - **Immunofluorescence mapping (IFM)**: Identifies the level of skin cleavage and detects abnormalities in structural proteins (e.g., keratins, laminin-332, type VII collagen). - **Transmission electron microscopy (TEM)**: Allows ultrastructural analysis of the dermal–epidermal junction to determine the exact plane of blister formation. - **Molecular genetic testing**: Confirms the diagnosis by identifying pathogenic variants in known EB-associated genes (e.g., *KRT5*, *KRT14*, *LAMB3*, *COL7A1*).

These laboratory methods are complementary and should be used sequentially or in parallel depending on clinical urgency and availability. Genetic testing is essential not only for definitive diagnosis but also for enabling prenatal testing, genetic counselling, and eligibility for emerging targeted therapies.

Early and accurate diagnosis facilitates appropriate multidisciplinary management, improves patient outcomes, and supports informed family planning.